ABSTRACT OF THE DISCLOSURE

A system and method for a high-radix computing architecture that replaces traditional binary (base-2) transistors with phototransistors capable of distinguishing between a plurality of discrete wavelengths of light. Each distinct wavelength, or color, corresponds to a unique logical state, enabling a base-N number system where N is greater than two. The system comprises an array of such phototransistors integrated into a processing unit, which is directly coupled to a massively parallel optical data bus. By processing data encoded as multiple distinct wavelengths, the system can achieve significantly higher computational density and power efficiency compared to conventional electronic, binary systems. This architecture eliminates the bottleneck of optical-to-electrical signal conversion and allows for computation at the native speed and parallelism of light.